Experimental Characterization of Bio-Based Roofing Composites with Natural Binder Matrix

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01 INTRODUCTION

The increasing demand for sustainable construction materials has driven research toward the development of eco-friendly composites derived from renewable resources. This study focuses on the experimental characterization of bio-based composites reinforced with treated agricultural fibers and bonded using industrial by-products. The composite is fabricated with the binder matrix and supplementary materials. The geopolymer matrix exhibits minimal shrinkage and creep , preventing structural deformations and long-term failures. Results indicate that the developed composite exhibits adequate structural performance, and improved thermal insulation compared to conventional roofing materials. The study demonstrates the potential of bio-based composites as sustainable roofing solutions, reduced environmental impact, and applicability in housing and rural infrastructure development.

02 OBJECTIVES

- ✓ To quantify the mechanical, thermal, and physical properties of the developed bio-based composite.
- ✓ To develop a sustainable bio-based roofing composite using treated agricultural fibers and an industrial by-product-based geopolymer binder matrix.
- ✓ To investigate the feasibility of low-cost roofing material for rural and affordable housing applications.

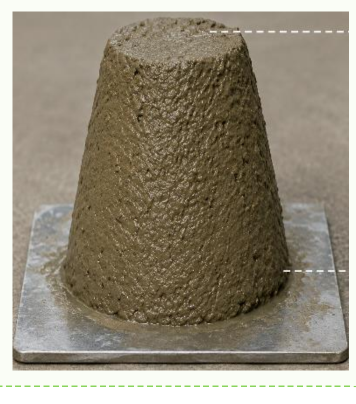
03 METHODOLOGY

Material Collection
- Binders — Alkaline Solutions
- Fine Aggregate — Banana Fiber
- Coarse Aggregate — Superplasticizer

Material Test
- Particle Size Distribution (PSD)
- SEM Analysis

Mix Design

Fresh Concrete Test
Slump Test



Preparation of Specimen
70.6 mm x 70.6 mm x 70.6 mm

Data Analysis & Results

04 RESULTS

Constituents	Quantity per kg/m³
Fly ash	73.5
GGBS	296
Fine Aggregate	850
Alkali Activator solution	257
Super Plasticizer	7.8
Banana Fibre	11.6

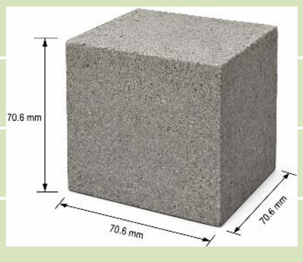


Table 1. Supporting evidence for the discussion

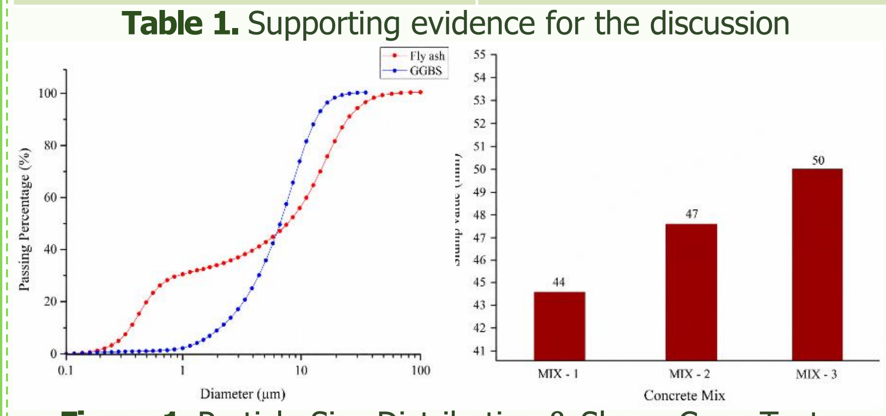


Figure 1. Particle Size Distribution & Slump Cone Test

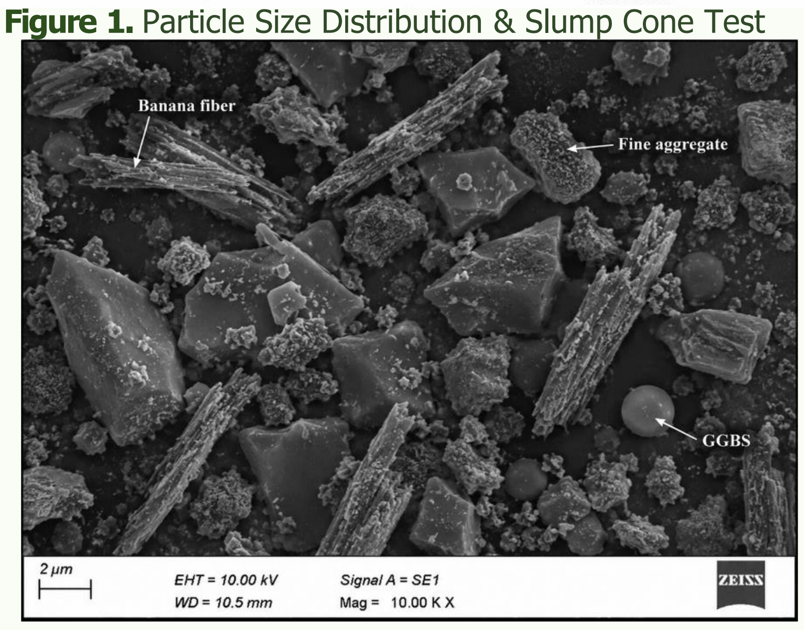
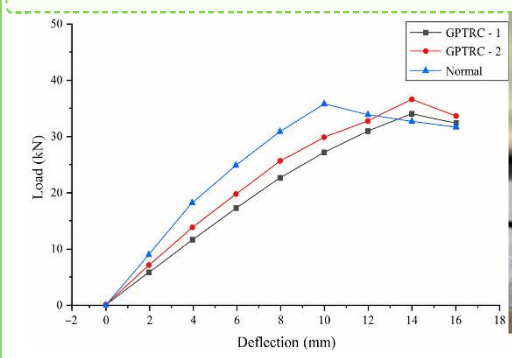


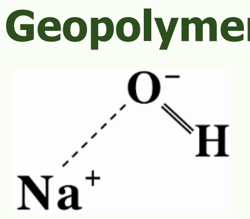
Figure 2. SEM Analysis



PARAMETER	R1	R2	R3
$NaOH:NaSiO_3$	1:2	1.94	1.95
NaOH (Molarity)	3	3.5	2.5
FA:GGBS	50:50	40:60	20:80

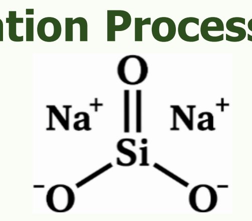
Table 2 Comparison Ratio of the chemical composition

Geopolymerisation Process :



Sodium Hydroxide

+



Sodium Silicate

=

Alkaline Activator

$n(SiO_2) + m(Al_2O_3) + Na_2O + H_2O$ — Alumino-silicate Source $\xrightarrow{\text{Alkaline Activation}}$ $Si(OH)_4 + Al(OH)_4^-$ (species in solution)

$\xrightarrow{\text{Polycondensation}}$ $Na\text{-}(Si\text{-}O\text{-}Al\text{-}O)_n\text{-}Si\text{-}O\text{-}Na + H_2O$ — Geopolymer Gel

05 DISCUSSION

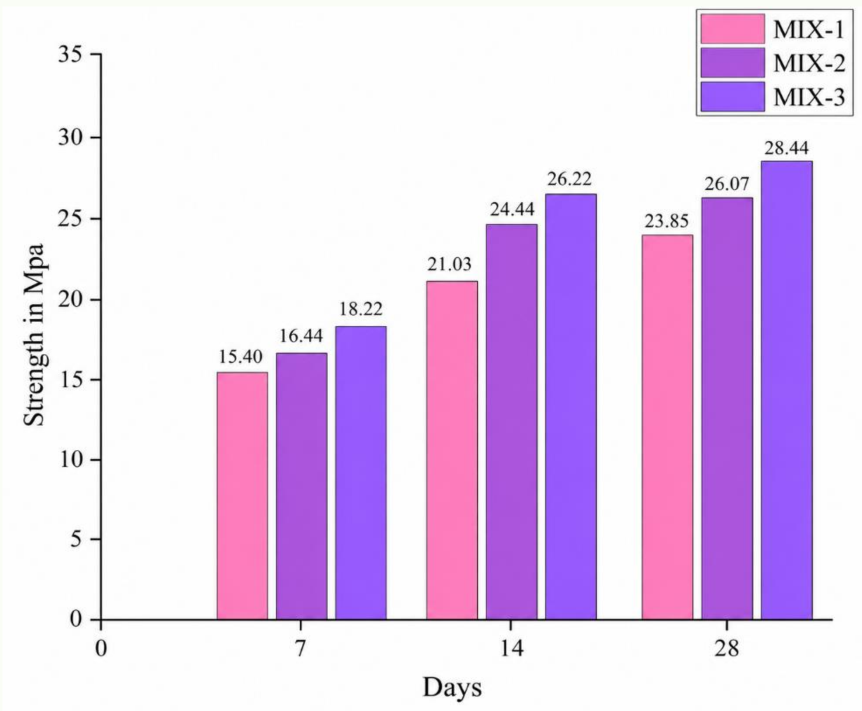


Figure 3. Compressive strength graph

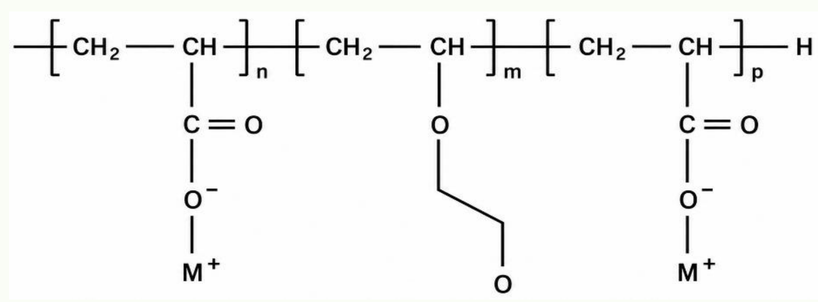


Figure 4. Super plasticizer polycarboxylate ether (PCE)

Typical Dosage: 0.5% - 1.5% by weight of binder (opt: 1.0%)

06 CONCLUSIONS

- ✓ Geopolymer Fibre incorporated mortar enhances mechanical properties.
- ✓ The use of Banana Fiber in the mix promotes sustainability by minimizing the carbon emissions.
- ✓ Third conclusion is relevant to circularity, energy or environment.

> *Valorization of agricultural residues through geopolymer technology enables durable, low-carbon roofing composites for next-generation sustainable infrastructure.*

07 FUTURE WORK

Optimization of the composite formulation, long-term durability evaluation, and large-scale field validation are essential for commercial implementation.

08 REFERENCES

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